

NIIS HACKATHON 2026 — OFFICIAL PROBLEM BRIEF

Track: AI & Machine Learning

Challenge ID: PS-08

Title: AI-Powered Workplace Safety Gear Detection

1. Executive Summary & Problem Overview

Occupational hazards in industrial manufacturing, heavy construction sites, warehousing, chemical refineries, and mining facilities account for thousands of preventable workplace injuries and fatalities annually. Compliance with mandatory Personal Protective Equipment (PPE) regulations is a critical legal and life-safety requirement.

Currently, workplace safety monitoring relies on periodic, manual physical inspections by safety supervisors or passive, unmonitored CCTV security feeds. Human oversight suffers from severe fatigue, blind spots across expansive worksites, delayed hazard intervention, and zero automated compliance logging. Workers frequently remove or forget critical protective equipment in hazardous zones without timely warning.

The objective of **PS-08 (AI-Powered Workplace Safety Gear Detection)** is to develop an edge-ready, computer vision-based real-time surveillance analytics system. The platform must process live CCTV video streams or recorded footage to detect workers and verify their compliance with mandatory safety equipment—including hardhats/helmets, high-visibility reflective vests, safety footwear, gloves, and eye/face shields—triggering instant real-time audio-visual alerts and automated safety violation incident logs.

2. Core Personas & Access Roles

1. **EHS (Environmental Health & Safety) Officer:** Reviews live multi-camera PPE compliance feeds, manages violation incident queues, evaluates site safety trends, and exports regulatory audit reports.
2. **Site Supervisor / Floor Manager:** Receives instantaneous floor-level violation push notifications to intervene immediately when unequipped personnel enter hazardous active machinery zones.
3. **Control Room Operator:** Monitors camera telemetry health, reviews flagged bounding-box snapshots, and verifies automated incident classification accuracy.
4. **Site Workers & Contractors:** Receive immediate audible or visual beacon feedback when entering a designated zone without mandatory protective gear.

3. Key Functional Modules & Technical Deliverables

Module 1: Multi-Class PPE Object Detection & Tracking

- **Person & Gear Localization:** Employs optimized deep learning object detection models to simultaneously localize workers and their respective PPE items (Hardhats/Helmets, High-Vis Reflective Vests, Safety Gloves, Protective Footwear, and Eye Goggles).
- **Body-Part Spatial Association:** Maps detected safety gear onto specific anatomical worker keypoints (e.g., verifying that a detected helmet is physically worn on the head rather than held in hand or sitting on a nearby table).
- **Multi-Camera Person Tracking:** Assigns unique persistent tracking IDs (e.g., ByteTrack, DeepSORT) across successive video frames to prevent duplicate alerts for the same walking individual.

Module 2: Dynamic Geofencing & Zone-Specific Safety Policies

- **Custom Restricted-Zone Mapping:** Allows administrators to define polygon geofences directly over camera views (e.g., High-Hazard Heavy Machinery Zone, Chemical Storage, General Walkway).
- **Granular Rule Engine:** Configures role- or zone-specific compliance policies (e.g., *"Zone A requires Helmets + High-Vis Vests; Zone B requires Helmets + Gloves + Protective Eye Shields"*).
- **Directional & Curfew Violations:** Flags unauthorized personnel movement into restricted perimeters during off-shift hours or unsafe proximity to operating excavators/forklifts.

Module 3: Real-Time Alerting & Multi-Channel Notification

- **Low-Latency Alarm Triggers:** Generates immediate alerts (<1 second from violation detection) via on-screen visual bounding box overlays (Green = Compliant, Red = Violation).
- **Automated Audio-Visual Warnings:** Triggers simulated siren beacons, PA speaker announcements, or digital signage boards at the entry point of the hazardous perimeter.
- **Mobile / Web Dispatch:** Delivers instant incident snapshots with timestamps, camera IDs, and missing gear tags to safety supervisors via WebSockets, Telegram, or SMS.

Module 4: Incident Analytics & Compliance Reporting

- **Digital Violation Logbook:** Stores tamper-evident incident records containing high-resolution violation snapshots, camera locations, timestamps, and missing gear classifications.
- **Safety Heatmaps & Trend Analytics:** Visualizes temporal patterns (e.g., peak violation hours, recurring non-compliant contractor teams, safety score trajectories over time).
- **Exportable Audit Dossiers:** Automated generation of weekly and monthly workplace safety compliance reports (PDF/CSV) ready for OSHA or regulatory labor compliance audits.

4. Recommended Technology Stack

Layer, Suggested Technologies

Frontend Web / Safety Console, React.js (Vite) / Next.js / Tailwind CSS / Canvas API (Bounding box overlays) / Recharts

Backend Architecture, Python (FastAPI / Flask) with WebSockets / WebRTC for live video streaming

Computer Vision & AI Models, "YOLOv8 / YOLOv9 / YOLOv11, OpenCV, PyTorch, Pretrained PPE datasets (Roboflow Universe, Construction Site Safety datasets), ByteTrack / DeepSORT"

Video Pipeline & Acceleration, "FFmpeg, RTSP streaming protocols, ONNX Runtime / TensorRT for edge inference acceleration"

Database & Storage, PostgreSQL / Supabase + MinIO / AWS S3 (for incident snapshot image storage)

Notification Services, "Redis Pub/Sub, WebSockets, Twilio / Telegram Bot API for supervisor alerts"

5. Evaluation & Judging Criteria (100 Points Total)

1. **Detection Accuracy & Model Precision (25 Points):** High mAP (mean Average Precision) across multiple PPE classes (Helmets, Vests, Gloves) under varied lighting, angles, and partial occlusions.
2. **Spatial Association & Logic Guardrails (25 Points):** Correctly associating equipment with worker anatomy and verifying zone-specific safety rules without generating excessive false positives.
3. **Real-Time Video Pipeline Performance (20 Points):** Smooth FPS processing (minimum 15–30 FPS on standard test video feeds/webcam) with low-latency alert dispatch.
4. **Dashboard UX & Incident Management (15 Points):** Clean safety monitoring console, intuitive geofence polygon configuration, and clear violation snapshot archives.
5. **Sprint Execution, Code & Demo Standards (15 Points):** Clean GitHub repository, provided sample test video footage, .env.example, and an interactive working prototype demonstration.

6. Submission Guidelines

1. **Code Repository:** Public/Private GitHub repository containing clean branch commits, setup instructions, sample test video clips (both compliant and violating footage), and .env.example.
2. **Live Deployment / Demonstration:** Hosted interactive web application with webcam or recorded video stream upload capabilities (Vercel / Render / Hugging Face Spaces / local screen recording).
3. **Demo Video / Deck:** A focused 3-minute video showing:
 4. Ingestion of CCTV / recorded video feed.
 5. Real-time detection and bounding-box tracking of workers and safety gear.

6. Dynamic violation trigger when a worker enters a geofenced zone without a helmet or vest.
7. Instant alert logging and export of the safety compliance audit report.